Daten:

$f(x,y) = \frac{x^2}{y}$ $\quad h = 0,7$

$x_0 = 0$ $\quad x_1 = 0,7$ $\quad x_2 = 1,4$

$y_0 = 2$ $\quad y(x) = \sqrt{\frac{2x^3}{3} + 4}$

Exakte:

$y(0) = \sqrt{\frac{2 \cdot 0^3}{3} + 4} = 0$

$y(0,7) = \sqrt{\frac{2 \cdot 0,7^3}{3} + 4} = 2,056372$

$y(1,4) = \sqrt{\frac{2 \cdot 1,4^3}{3} + 4} = 2,414401$

a)

$y_{i+1} = y_i + h \cdot f(x_i, y_i)$

Schritt 1

$f(x_1, y_1) = f(0,7, 2) = \frac{0,7^2}{2} = \frac{0,49}{2} = 0,245$

$y_1 = 2 + 0,7 \cdot 0,245 = 2,1715$

Fehler

$|y(0) - y_0| = |2 - 2| = 0$

$|y(0,7) - y_1| = |2,056372 - 2| = 0,056372$

$|y(1,4) - y_2| = |2,414401 - 2,1715| = 0,242901$

b)

$k_1 = f(x_i, y_i)$

$x_{i+\frac{1}{2}} = x_i + \frac{h}{2}$

$y_{i+\frac{1}{2}} = y_i + \frac{h}{2} \cdot k_1$

$y_{i+1} = y_i + h \cdot f(x_{i+\frac{1}{2}}, y_{i+\frac{1}{2}})$

Schritt 1

$k_1 = f(0,2) = 0$

$x_{\frac{1}{2}} = 0 + 0,35 = 0,35$

$y_{\frac{1}{2}} = 2 + 0,3 \cdot 0 = 2$

$f(0,35; 2) = \frac{0,35^2}{2} = \frac{0,1225}{2} = 0,06125$

$y_1 = 2 + 0,7 \cdot 0,06125 = 2,042875$

Schritt 2

$k_1 = f(0,7; 2,042875) = \frac{0,49}{2,042875} = 0,239858$

$x_{1+\frac{1}{2}} = 0,7 + 0,35 = 1,05$

$y_{1+\frac{1}{2}} = 2,042875 + 0,35 \cdot 0,239858$

$y_{1+\frac{1}{2}} \approx 2,126825$

$f(1,05; 2,126825) = \frac{1,05^2}{2,126825} = \frac{1,1025}{2,126825} \approx 0,518378$

$y_2 = 2,042875 + 0,7 \cdot 0,518378 = 2,40576$

Fehler

$|y(0) - y_0| = 0$

$|y(0,7) - y_1| = |2,056772 - 2,042875| = 0,013497$

$|y(1,4) - y_2| = |2,414401 - 2,405740| = 0,008601$

c) $k_1 = f(x_i, y_i)$ $\qquad \tilde{y}_{i+1} = y_i + h \cdot k_1$

$k_2 = f(x_{i+1} ; \tilde{y}_{i+1})$ $\qquad y_{i+1} = y_i + \frac{h}{2}(k_1 + k_2)$

Schritt 1

$k_1 = f(0,2) = 0$

$\tilde{y}_1 = 2 + 0{,}7 \cdot 0 = 2$

$k_2 = f(0{,}7, 2) = \frac{0{,}42}{2} = 0{,}245$

$y_1 = 2 + \frac{0{,}7}{2}(0 + 0{,}245$

$y_1 = 2 + 0{,}35 \cdot 0{,}245 = 2{,}0875$

Schritt 2

$k_1 = f(0{,}7 ; 2{,}08575) = \frac{0{,}49}{2{,}08575} \approx 0{,}234927$

$\tilde{y}_2 = 2{,}08575 + 0{,}7 \cdot 0{,}234927 \approx 2{,}250199$

$y_2 = 2{,}08575 + \frac{0{,}7}{2}(0{,}234927 + 0{,}871036)$

$= 2{,}08575 + 0{,}35 \cdot 1{,}105961 \approx 2{,}472837$

Fehler

$|y(0) - y_0| = 0$

$|y(0{,}7 - y_1| = |2{,}056372 - 2{,}08575| = 0{,}029378$

$|y(1{,}4) - y_2| = |2{,}414401 - 2{,}472837| = 0{,}058635$